

Hamra Coffee Shop – Data Analytics Project

Sales, Customers & Performance Insights (2024)
Wael Saleh | 20/11/2025

Objective:

Analyze 2024 sales & customer data from a Lebanese-style café in Hamra using Python, Pandas, and Tableau.

Goal:

Create a clean dataset → extract insights → build a strategic dashboard for management decisions.

Datasets Used:

- Sales (5,000 rows)
- Products (53 items)
- Customers (200 profiles)
- Time dimension (full year)

Workflow Summary

1. Data Loading (Python, Pandas)

2. Data Cleaning & Validation

- Fix dates
- Ensure consistent Product_ID & Customer_ID
- Remove duplicates
- Create Total_Sale & Profit
- Add Age Groups

3. Merging Datasets

4. Exploratory Data Analysis (EDA)

5. Tableau Dashboard

KPI Metrics

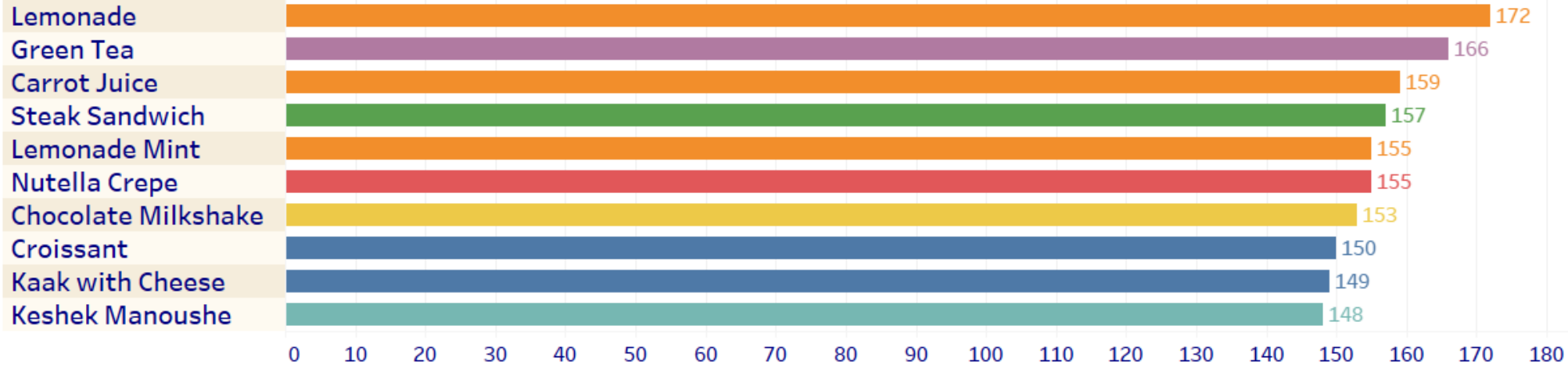
Quantity	1,564
Total Cost	1,406
Total Profit	4,591
Total Revenue	6,651
Total Orders	4,991
Items Per Order	1.44
AOV	6.12

- The café shows strong yearly performance with a total revenue of **\$31,972**, profit of **\$21,883**, and **7,018 items** sold across **4,991 orders**.
- The **Average Order Value (AOV)** is **\$6.41**, and each customer buys **1.41 items per order**, reflecting affordable pricing and frequent, low-ticket purchases.

These KPIs form the basis of performance evaluation for the café

Top 10 Products – Quantity Sold

Product Name



Category

(All)

Product Name

(All)

Limit

Top 10 by
SUM([Quantity])

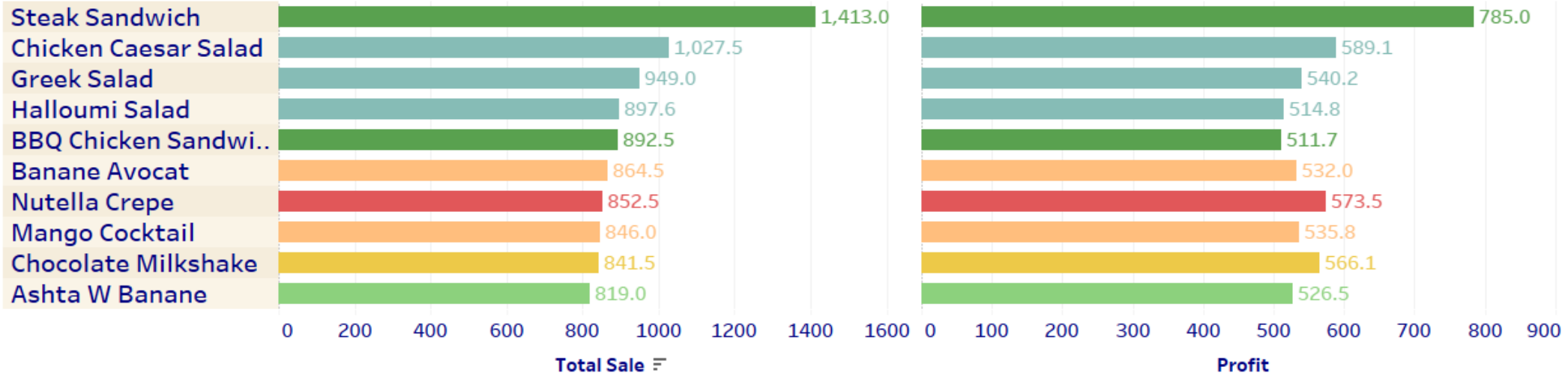
Category

- Bakery
- Cold Drinks
- Desserts
- Manakish
- Sandwiches
- Shakes
- Tea

- **Lemonade (172 orders)** is the most purchased item, showing strong demand for refreshing cold drinks.
 - **Green Tea (166)** and **Carrot Juice (159)** highlight a preference for lighter, health-focused beverages.
 - **Steak Sandwich (157)** leads the savory category as the top food item.
 - **Nutella Crepe (155)** and **Chocolate Milkshake (153)** perform extremely well in the sweet category.
 - Popular bakery items include **Croissant (150)** and Lebanese classics such as **Kaak with Cheese (149)** and **Keshek Manoushe (148)**.
- Cold drinks dominate volume, while sandwiches and desserts show strong, balanced demand.

Top 10 Products – Revenue & Profit

Product Name



Product Name

(All)

Limit

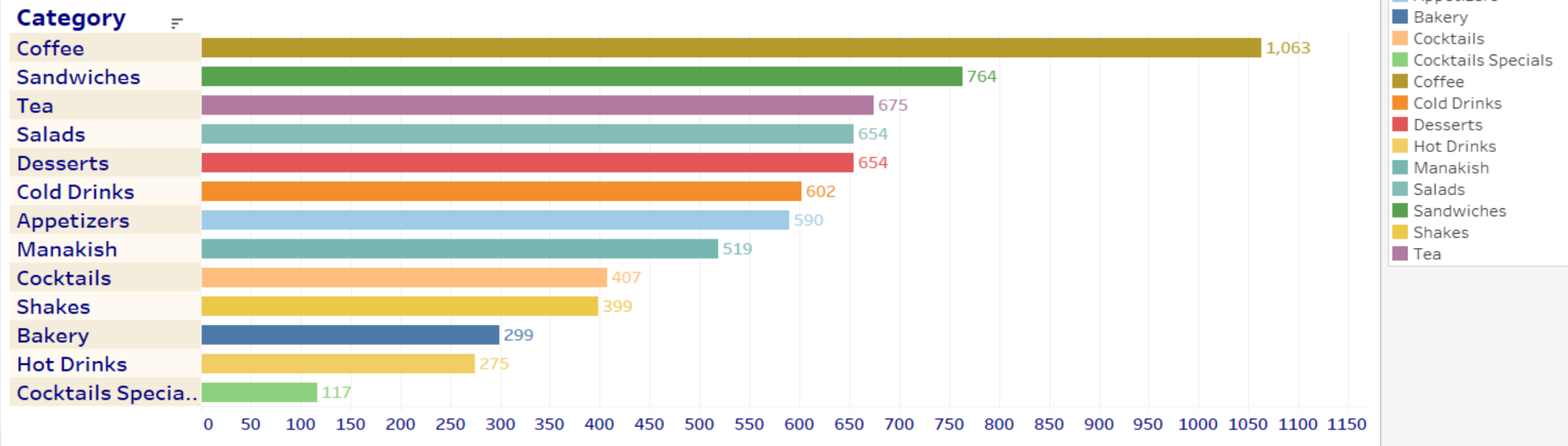
Top 10 by SUM([Total Sale])

Category

- Cocktails
- Cocktails Specials
- Desserts
- Salads
- Sandwiches
- Shakes

- **Steak Sandwich** dominates both revenue and profit, making it the café's most valuable product overall.
- Salads such as **Chicken Caesar** and **Greek Salad** consistently rank high in revenue and profit, showing strong demand from health-conscious customers.
- Desserts and shakes—including **Nutella Crepe**, **Chocolate Milkshake**, and **Mango Cocktail**—stand out with high profitability due to low preparation costs and steady sales volume.
- Sweet and specialty beverages contribute significantly to both revenue and profit, confirming their importance in customer choice patterns.
- → *Overall: Sandwiches and salads drive financial performance, while desserts and cocktails provide high-margin growth opportunities.*

Top Categories – Quantity Sold

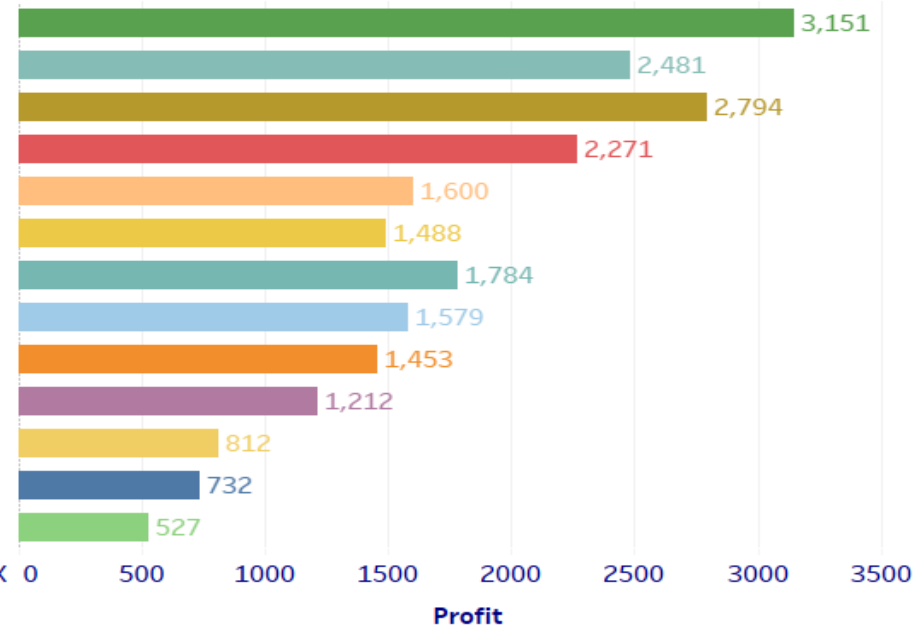
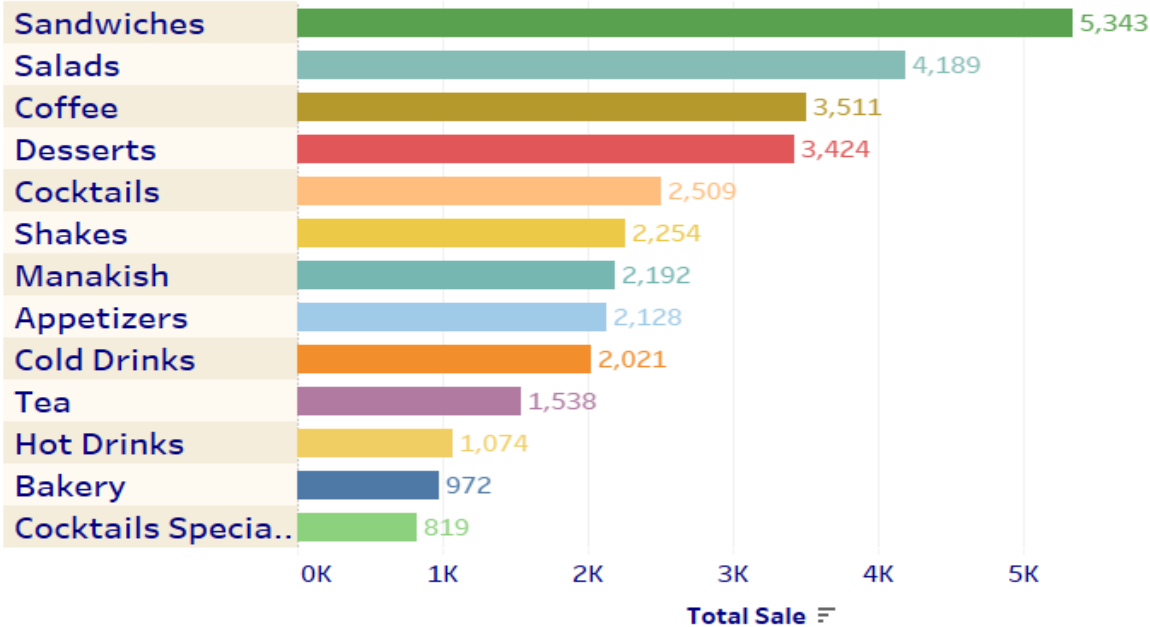


- The **Coffee** category leads in total units sold, confirming its role as the café's daily essential and most frequent purchase driver.
- **Sandwiches** and **Tea** also show strong volume, reflecting consistent customer demand across meal and beverage options.
- **Desserts**, **Salads**, and **Cold Drinks** maintain balanced popularity throughout the year. Traditional Lebanese items such as **Manakish** and **Bakery products** continue to perform well, preserving cultural appeal among regular customers.

→ *Overall: Coffee drives volume, while sandwiches, tea, and desserts ensure steady all-day demand.*

Top Categories – Revenue & Profit

Category



Product Name

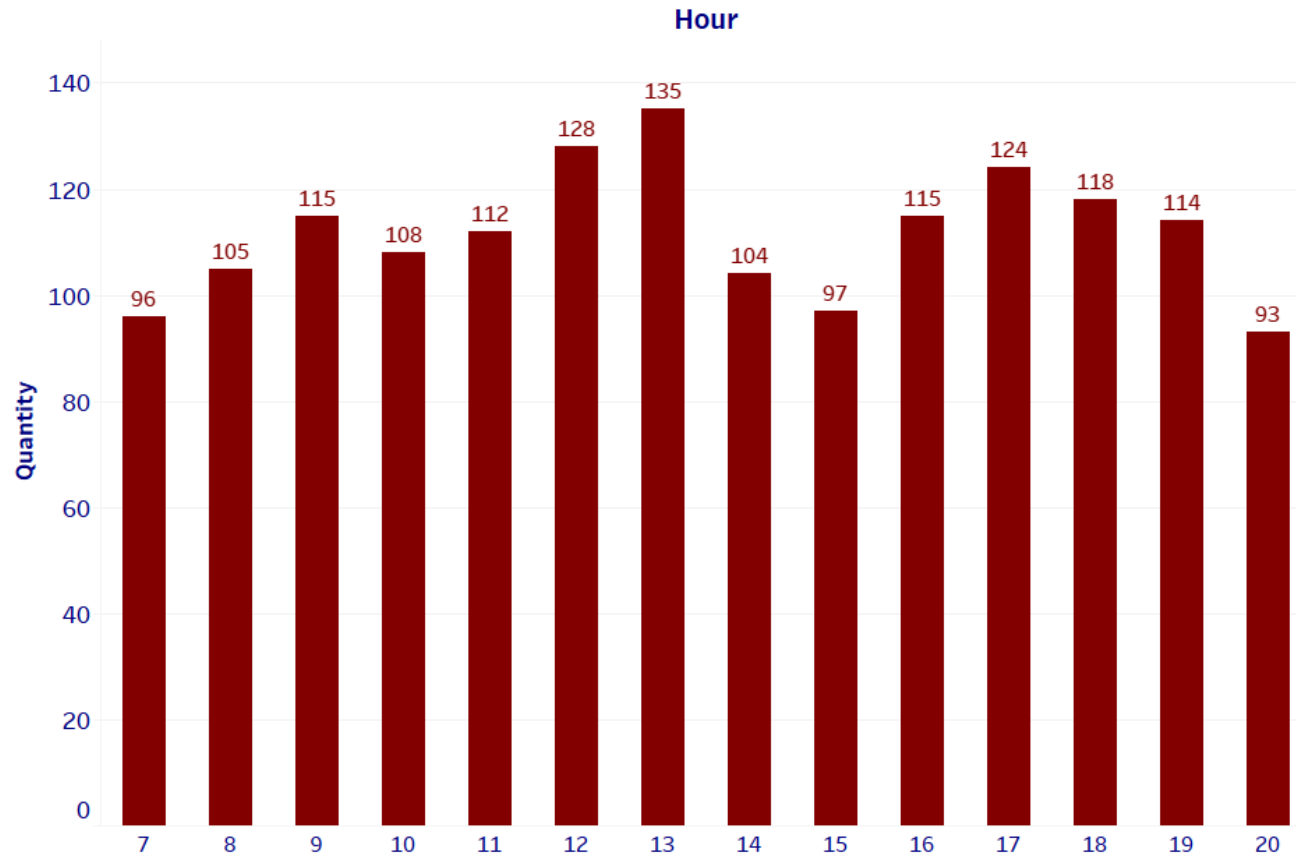
(All)

Category

- Appetizers
- Bakery
- Cocktails
- Cocktails Specials
- Coffee
- Cold Drinks
- Desserts
- Hot Drinks
- Manakish
- Salads
- Sandwiches
- Shakes
- Tea

- **Sandwiches** generate the highest revenue and profit, making them the café's strongest financial pillar.
 - **Salads** and **Coffee** also provide excellent returns, combining strong demand with healthy profit margins.
 - **Desserts**, **Cocktails**, and **Shakes** deliver substantial revenue and maintain high profitability thanks to low ingredient costs.
 - Regional favorites like **Manakish** offer consistent profitability, while **Cold Drinks** add seasonal revenue growth.
- Overall: Sandwiches lead financially, while salads, coffee, desserts, and cocktails provide reliable high-margin performance.

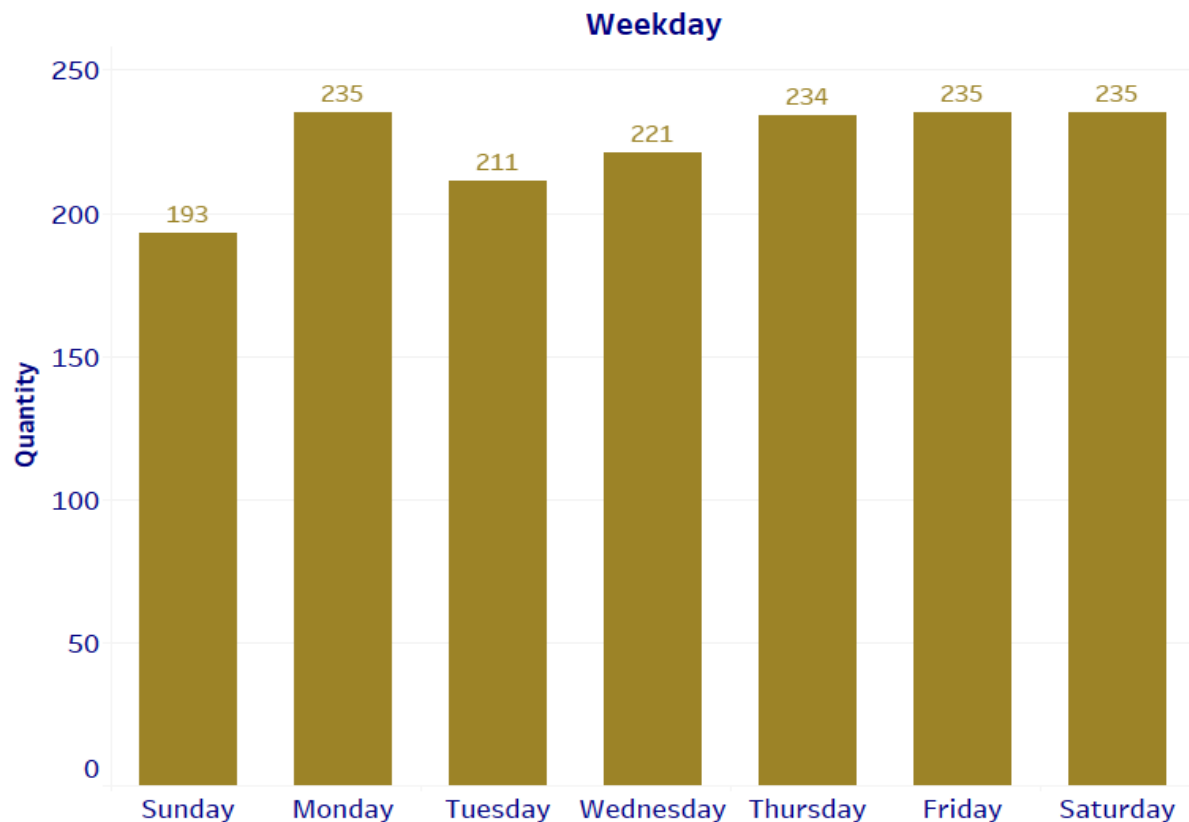
Sales by Hour



- Customer activity follows clear daily patterns.
- The busiest times are **1 PM**, **4 PM**, and **7 PM**, corresponding to lunch, afternoon breaks, and early dinner periods.
- Morning hours (7–10 AM) show steady traffic driven mostly by coffee purchases.
- Sales gradually decrease after 9 PM, reflecting typical café behavior and reduced nighttime demand.

→ *Peak hours offer strong opportunities for meal combos, dessert upsells, and targeted promotions.*

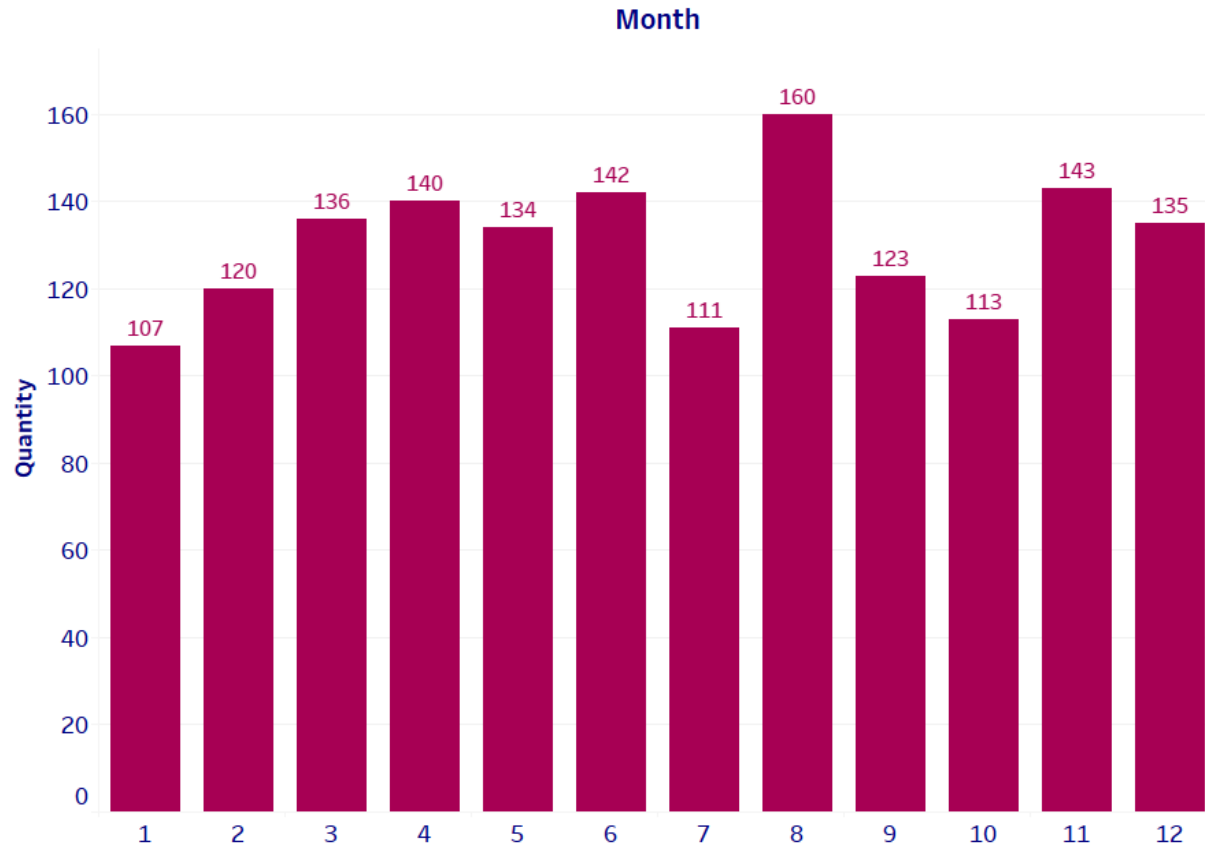
Sales by Weekday



- Sales peak on **Saturday**, making it the strongest day of the week.
- **Friday and Sunday** also show high volume, indicating strong weekend-driven customer behavior.
- Monday through Thursday remain stable but slightly lower, with **Thursday showing the weakest activity**.
- This pattern suggests that customers treat the café as a weekend destination for meals, gatherings, and leisure.

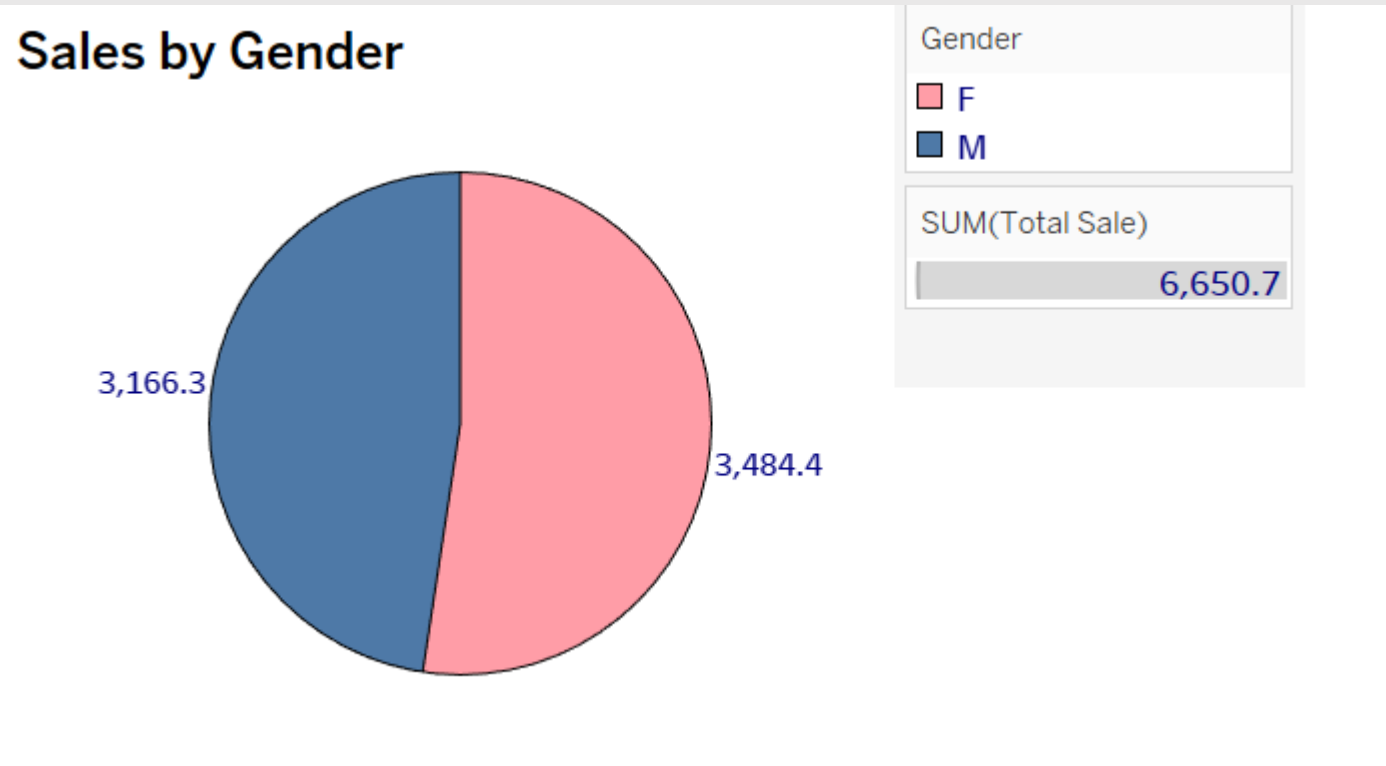
→ *Weekend-focused offers and special menus can boost revenue even further.*

Monthly Revenue



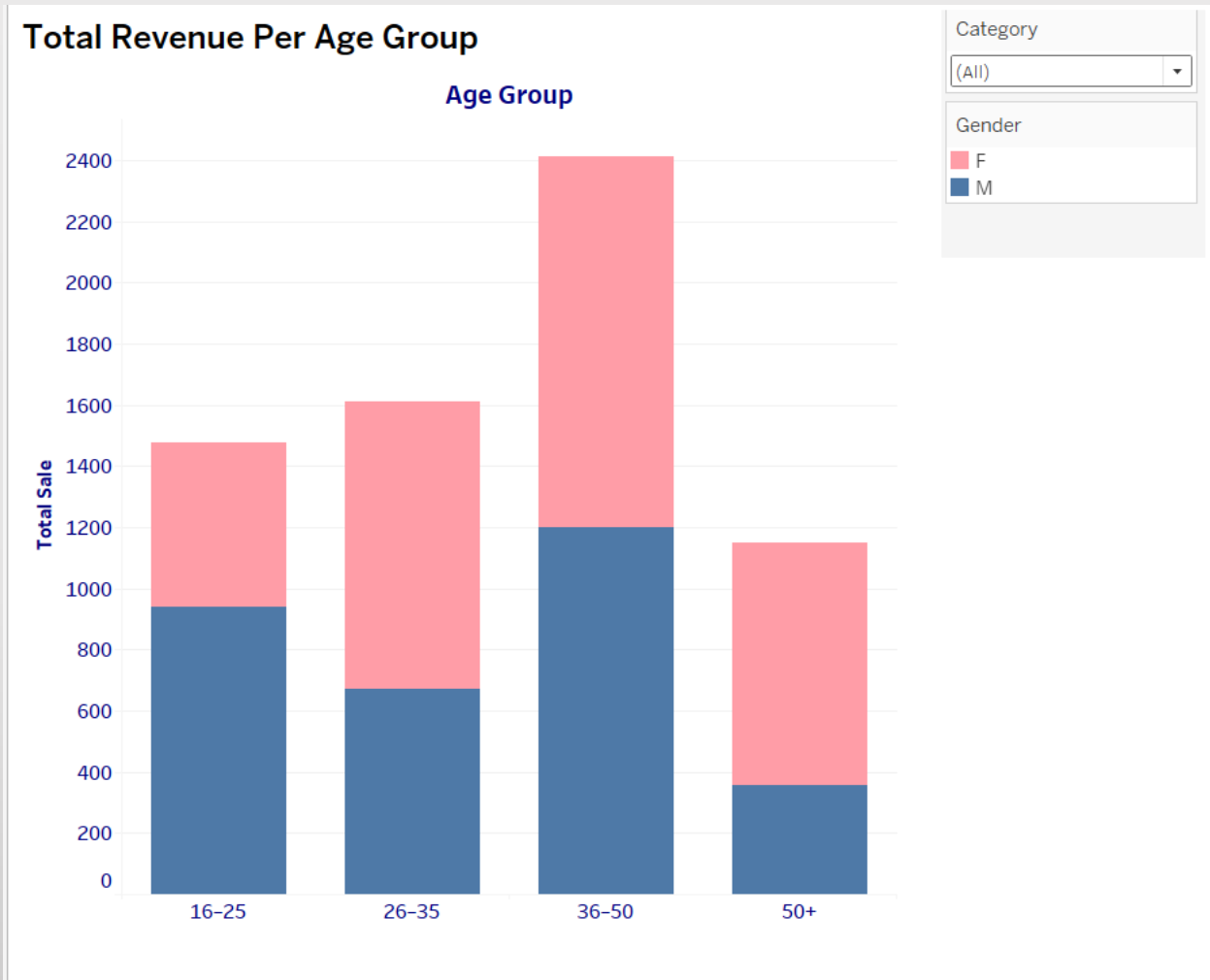
- **August** shows the highest sales peak of the year, indicating a strong surge in customer demand during late summer.
- **March, April, June, and November** also perform strongly, each maintaining quantities above 135 orders.
- The beginning of the year starts slower in **January**, with steady growth through **February** into mid-year.
- A noticeable dip appears in **July** and **October**, showing slightly reduced customer activity before and after the summer peak.

→ The café experiences a strong late-summer boost, with opportunities to reinforce sales during slower mid-summer and early-autumn months.



- Female customers contribute slightly more purchases and spending.
- Women strongly prefer salads, desserts, and tea, while men buy more sandwiches, coffee, and appetizers.
- Both genders share similar preferences for cold drinks and daily coffee.

→ *Gender-tailored promotions can improve customer engagement.*



- Ages **36–50** represent the café's most valuable demographic, with the highest spending and order counts.
- Younger customers (16–25) prefer drinks and desserts, especially sweet and refreshing items.
- Ages **26–35** are balanced buyers across all categories.
- The **50+** group gravitates toward warm drinks, salads, and simple meals.

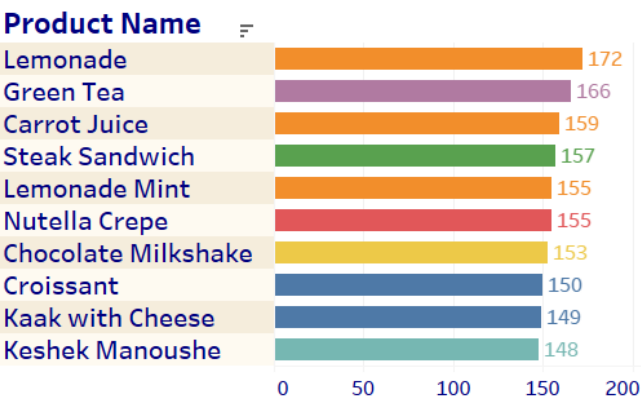
→ *Multi-segment targeting enhances reach; each age group responds to different menu categories.*

KPI Metrics

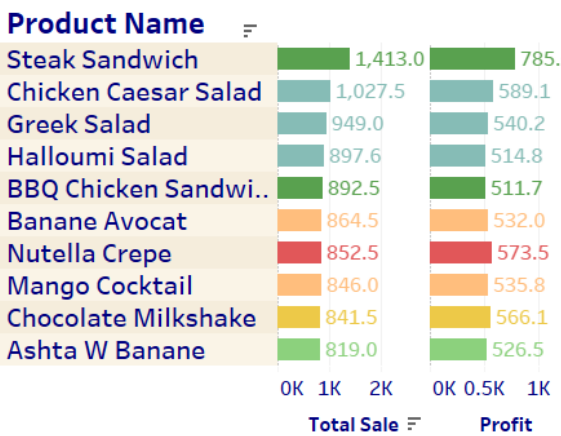
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Top 10 Products – Quantity Sold



Top 10 Products – Revenue & Profit



Category
(All)

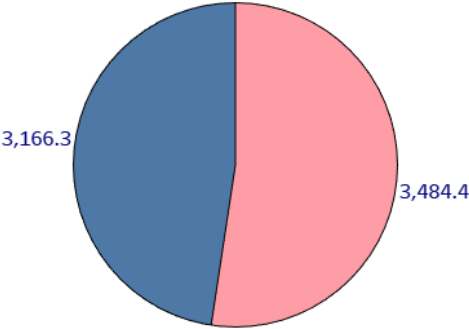
Product Name
(All)

Gender

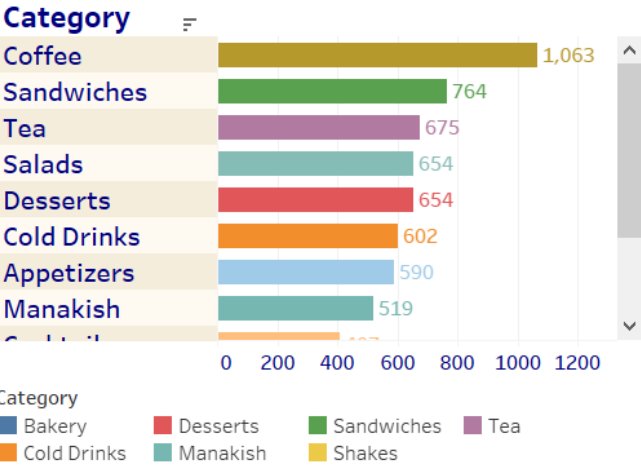
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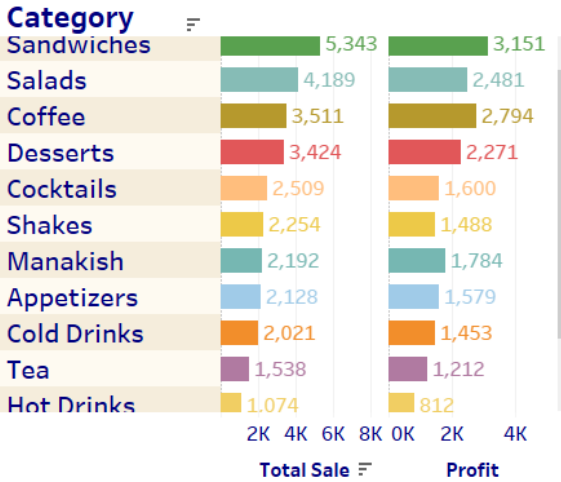
Sales by Gender



Top Categories – Quantity Sold



Top Categories – Revenue & Profit



Insights:

- Sales are consistently strong across the year, with **August and November** showing peak performance.
- **Cold Drinks, Sandwiches, and Desserts** lead in both quantity sold and revenue generated.
- Customer behavior is simple and steady — **most orders contain only one item**, keeping AOV around **\$6.40**.
- The café achieves a healthy **profit margin**, with many high-performing signature items (e.g., Steak Sandwich, Nutella Crepe).
- The **36–50 age group** and **female customers** contribute the highest sales across multiple categories.
- Product demand shows clear patterns, providing strong support for forecasting and menu decisions.
- Overall, the café operates with **stable demand, strong category leaders, and reliable customer segments**.

Recommendations:

- Expand top-performing categories (**Cold Drinks, Sandwiches, Desserts**) with new variations and seasonal items.
- Use **bundle deals, combos, and upsell prompts** to increase Items Per Order (IPO = 0.31).
- Focus marketing efforts on **female customers** and the **36–50 age group**, who make up the most profitable segment.
- Launch **seasonal promotions** during high-demand months (August & November) to maximize revenue peaks.
- Introduce loyalty programs or digital punch cards to increase repeat visits and spending.
- Optimize menu layout by highlighting high-margin best-sellers to guide customer choices.
- Attract younger customers (16–25) with student offers, trending drinks, or social-media-driven promotions.
- Use monthly, daily, and hourly sales patterns to align staffing, inventory, and production schedules more efficiently.